

ST07

Biotechnology and Genetic Engineering

Biotechnology supplies 2-4 questions in the Science block and feeds Current Affairs on top. Biofuel generations were asked in 2013 and again in 2024. Bt cotton, GEAC and the COVID vaccine platforms are the other recurring anchors. Tools - restriction enzymes, vectors, PCR - are asked as straight one-liners.

Biotechnology is where the science paper and the current-affairs paper overlap. Half of this chapter is settled textbook material - restriction enzymes, PCR, tissue culture - and the other half moves every year: CRISPR therapies, GM mustard in the Supreme Court, ethanol blending targets, new vaccine platforms. Learn the fixed half as tables and keep the moving half dated.

Biotechnology

The toolkit

- Restriction enzyme = molecular scissors
- DNA ligase = molecular glue
- Vector = carrier (plasmid, phage, cosmid)
- Host = *E. coli*, *Agrobacterium*

Techniques

- PCR - Kary Mullis, Taq polymerase
- 94 C denature, 50-55 C anneal, 72 C extend
- Gel electrophoresis - DNA moves to anode
- Southern DNA, Northern RNA, Western protein

Gene editing

- CRISPR-Cas9, guide RNA
- Charpentier and Doudna, Nobel 2020
- FELUDA test - CSIR-IGIB
- Casgevy - first approved CRISPR therapy

GM crops in India

- Bt cotton 2002 - the ONLY approved GM crop
- Bt brinjal - moratorium 2010
- GM mustard DMH-11 - SC split verdict 2024
- GEAC is the apex approver

Plant biotech

- Totipotency, Haberlandt
- Explant - callus - plantlet, MS medium
- Auxin high = roots, cytokinin high = shoots
- Meristem culture = virus-free plants

Biofuels

- 1G edible crops
- 2G lignocellulosic residue - Panipat
- 3G algae
- 4G engineered algae, carbon negative
- E20 achieved 2025

Medicine

- Humulin - first rDNA product
- Dolly 1996, SCNT
- iPSC - Yamanaka and Gurdon 2012
- Gene therapy 1990, ADA deficiency

Institutions

- DBT 1986, BIRAC 2012
- GEAC under MoEFCC
- NBA Chennai, Biological Diversity Act 2002
- BioE3 Policy 2024

What genetic engineering is - and the toolkit it runs on

Biotechnology is the use of living organisms or their parts to make a product or a service. That definition is old - curd, bread and beer are all biotechnology. What is new is genetic engineering: deliberately cutting a chosen gene out of one organism and putting it into another, so that the second organism now makes something it never made before. To do that you need four things - scissors to cut the DNA, glue to join it, a carrier to take it inside a cell, and a host cell to multiply it. Learn the four roles first and every technique in this chapter falls into place.

- Recombinant DNA technology was developed by Stanley Cohen and Herbert Boyer in 1972-73.
- Paul Berg made the first recombinant DNA molecule and received the Nobel Prize in Chemistry in 1980.
- Restriction endonucleases are the molecular scissors; DNA ligase is the molecular glue.
- The first restriction enzyme isolated was Hind II. Type II restriction enzymes cut palindromic sequences and are the ones used in genetic engineering.
- The first commercial recombinant DNA product was human insulin - Humulin, 1982-83.

The four tools and what each one does

Tool	Role	Example
Restriction endonuclease	Cuts DNA at a specific sequence	EcoRI, Hind II, BamHI
DNA ligase	Seals the cut ends together	T4 DNA ligase
Vector	Carries the gene into the host cell	pBR322, lambda phage, Ti plasmid
Host cell	Multiplies the recombinant DNA and expresses it	E. coli, yeast, plant or animal cell

Restriction enzymes, vectors and plasmids

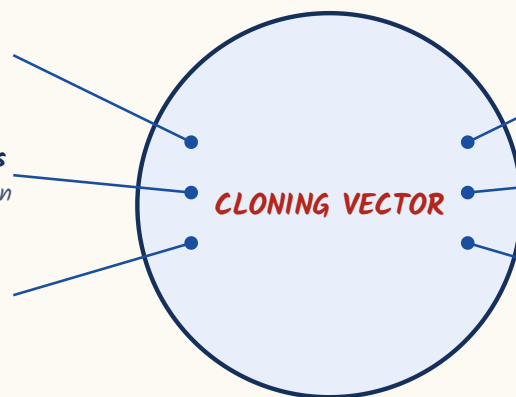
A restriction enzyme does not cut DNA at random. It recognises a specific short sequence - usually a palindrome, reading the same on both strands in the 5' to 3' direction - and cuts within it. If the cut is staggered rather than straight across, the two fragments are left with short single-stranded overhangs. Those are the sticky or cohesive ends, and they are the whole trick of the technology: DNA from two completely different organisms, cut by the same enzyme, will carry complementary overhangs and will pair up. A vector is simply the vehicle that carries the joined DNA into a living cell, and it must satisfy three conditions before it can do the job.

What every cloning vector must carry

ori
origin of replication

Unique restriction sites
one place to drop the gene in

Ti plasmid
the vector for plant cells



Selectable marker
usually antibiotic resistance

pBR322
first widely used plasmid

Lambda, cosmid, BAC, YAC
for larger inserts

- *EcoRI* is named from *Escherichia coli* strain RY13 - the first letter of the genus, the first two of the species, then the strain.
- *EcoRI* recognises GAATTC and cuts between G and A, leaving sticky ends.
- A cloning vector needs three things - an origin of replication (*ori*) so it can copy itself, and a selectable marker so transformed cells can be picked out.
- The third requirement is unique restriction sites, so the gene can be inserted at a known place.
- Gene transfer into animal cells - microinjection; into plant cells - biolistics or the gene gun, and *Agrobacterium*-mediated transfer.
- Other methods - electroporation, liposome-mediated transfer, and treating bacteria with divalent calcium to make them competent.

Vectors you should be able to name

Vector	What it is	Note
pBR322	Artificial bacterial plasmid	The first widely used plasmid vector
pUC19	Artificial plasmid	Improved cloning vector
Bacteriophage lambda	Virus that infects bacteria	Takes larger inserts than a plasmid
Cosmid	Plasmid plus phage packaging sites	Larger inserts still
BAC and YAC	Bacterial and yeast artificial chromosomes	Very large inserts, used in genome projects
Ti plasmid	Tumour-inducing plasmid of <i>Agrobacterium tumefaciens</i>	The commonest vector for dicot plants

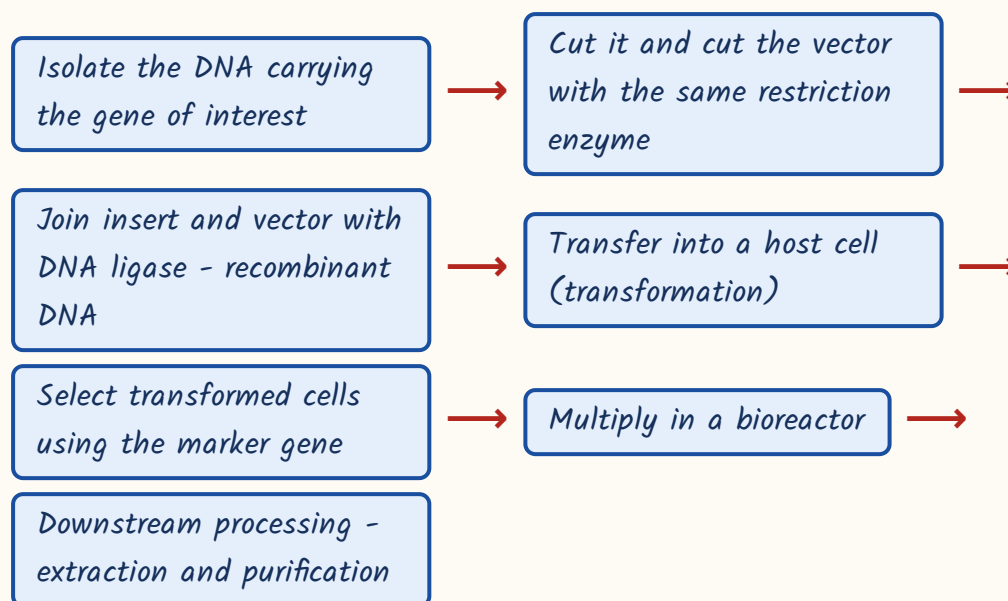
TRAP

Agrobacterium tumefaciens is a plant pathogen - it causes crown gall disease - and yet it is called the natural genetic engineer of plants, because in nature it already transfers its own DNA into the plant genome. Scientists simply disarmed it and loaded their own gene. Do not treat 'pathogen' and 'vector' as contradictory here.

Recombinant DNA technology - the sequence, step by step

This is the process the whole chapter is built on, and it is a sequence, so it can be asked as an ordering question. Say it as a story. You isolate the DNA that carries your gene. You cut it with a restriction enzyme, and cut the vector with the same enzyme so the ends match. You join the two with ligase, which gives you recombinant DNA. You push that into a host cell. The host divides, and every daughter cell carries your gene. Then you select the cells that actually took it up, grow them in a bioreactor, and finally purify the product downstream.

The rDNA sequence - learn it in order



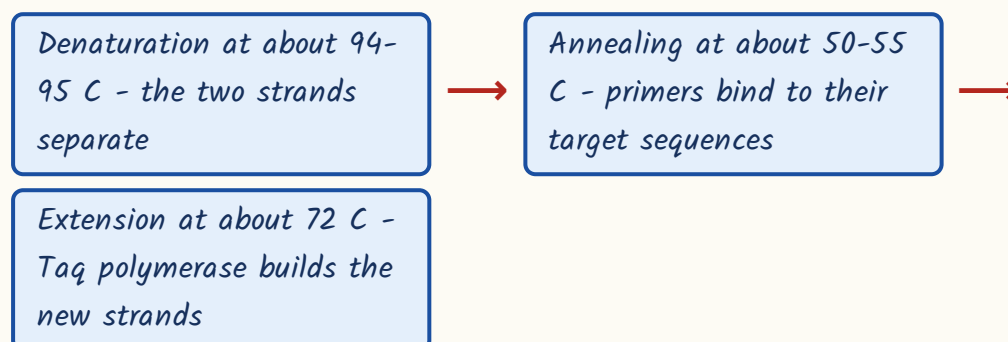
- Bacteria are made 'competent' to take up DNA by treatment with divalent calcium ions followed by a brief heat shock.
- The selectable marker - classically antibiotic resistance - lets you kill off every cell that did not take up the vector.
- A bioreactor provides the controlled temperature, pH, oxygen and stirring needed for large-scale culture; the stirred-tank type is the commonest.
- Downstream processing means separation, purification, formulation and quality control - it is counted as part of the technology, not an afterthought.

- Human insulin (Humulin) was the first rDNA product to be marketed, in 1982-83.
- The A chain (21 amino acids) and the B chain (30 amino acids) were produced separately in *E. coli* and then joined by disulphide bonds.
- Before this, diabetics used insulin extracted from pig and cattle pancreas, which caused allergic reactions in some patients.
- Other rDNA products in use - human growth hormone, erythropoietin, clotting factor VIII, hepatitis B vaccine.

PCR and gel electrophoresis - the two workhorse techniques

PCR - the polymerase chain reaction - is a photocopier for DNA. From a trace amount you get millions of copies in a couple of hours, which is why it underlies forensics, diagnosis and every cloning experiment. It works by repeating a three-step temperature cycle, and the enzyme that made it practical is Taq polymerase from a hot-spring bacterium, because an ordinary polymerase would be destroyed by the denaturation heat. Gel electrophoresis is the other half of the pair - it sorts the DNA fragments you now have, by size.

One PCR cycle - repeat it 25-35 times



- PCR was invented by Kary Mullis, who won the Nobel Prize in Chemistry in 1993.
- Taq polymerase comes from *Thermus aquaticus*, a bacterium of hot springs - it survives the 94 C step.
- Each cycle doubles the DNA, so the growth is exponential.
- RT-PCR - reverse transcription PCR - is the gold standard test for RNA viruses, including SARS-CoV-2. It first converts viral RNA into DNA.
- In agarose gel electrophoresis, DNA is negatively charged because of its phosphate backbone, so it moves towards the anode - the positive electrode. Smaller fragments travel further.
- Bands are stained with ethidium bromide and viewed under UV light.

The blotting techniques - a standard matching set

Technique	Molecule detected
Southern blot	DNA
Northern blot	RNA
Western blot	Protein

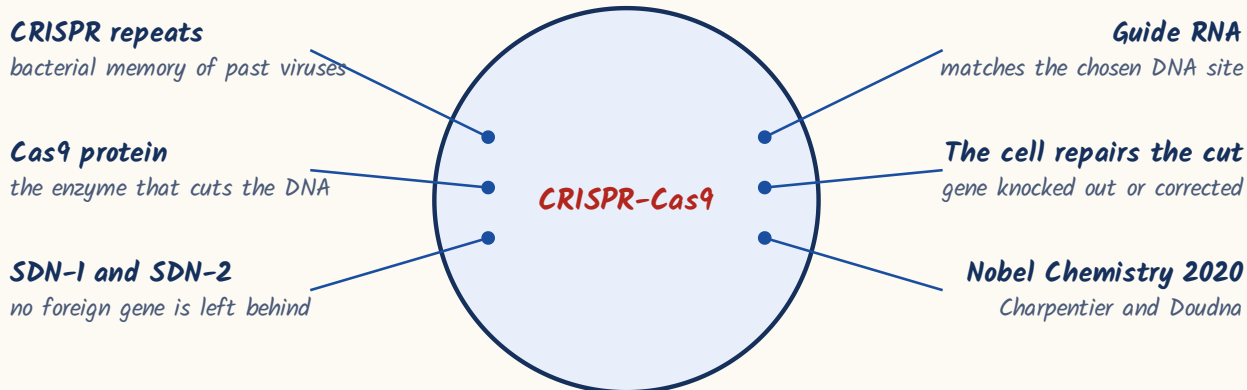
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Blots: Southern is the only one named after a person, Edwin Southern, and it is for DNA. The others were named as a joke on compass directions. Order them alphabetically and by molecule - DNA, RNA, protein for Southern, Northern, Western. And in electrophoresis remember: DNA is negative, so it runs to the positive anode.

CRISPR-Cas9 and genome editing

CRISPR-Cas9 is not a way of adding a foreign gene. It is a way of editing the organism's own DNA at a chosen spot - correcting a letter, knocking a gene out, switching one off. Bacteria evolved it as an immune memory against viruses: they store snippets of viral DNA between repeated sequences, and use those snippets as a guide to cut that virus if it returns. Scientists borrowed the system. A short guide RNA carries the Cas9 protein to the exact matching sequence, and Cas9 makes the cut. Everything after that is the cell's own repair machinery. Because the edit leaves no foreign gene behind, regulators treat some kinds of editing differently from GM - and India has used that distinction.

How CRISPR-Cas9 works



- CRISPR stands for Clustered Regularly Interspaced Short Palindromic Repeats; Cas9 is CRISPR-associated protein 9.
- Emmanuelle Charpentier and Jennifer Doudna won the 2020 Nobel Prize in Chemistry for it.
- FELUDA, a CRISPR-Cas9 based paper-strip test for COVID-19, was developed by CSIR-IGIB, New Delhi.
- Casgevy became the first approved CRISPR gene-editing therapy, for sickle cell disease and beta thalassaemia.
- SDN-1 and SDN-2 edits, which introduce no foreign DNA, are exempted from India's GMO rules; SDN-3, which inserts a foreign gene, is not.

Genetic modification (GMO) vs Genome editing

Genetic modification - GMO	Genome editing - CRISPR SDN-1/SDN-2
A foreign gene from another species is inserted	The organism's own DNA is altered; no foreign gene remains
Insertion site is largely random	The change is made at a chosen, precise site
Example - Bt cotton carrying a bacterial cry gene	Example - DRR Dhan 100 'Kamala' rice
Needs full GEAC clearance under the 1989 Rules	SDN-1 and SDN-2 exempted from GMO rules in India
Product is legally a GMO	Product is treated like a conventionally bred variety

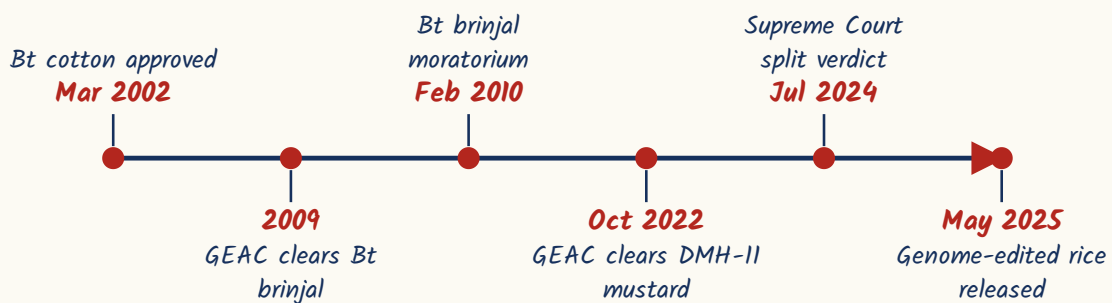
CURRENT LINK

Current link (as of September 2026): India released its first genome-edited rice varieties in May 2025 - DRR Dhan 100 'Kamala' from ICAR-IIRR and Pusa DST Rice 1 from ICAR-IARI, both made with SDN-1/SDN-2 editing and therefore outside the GMO rules. This is the fact that shows why the GMO-versus-genome-editing distinction matters in practice, and it is fresh enough to be asked.

Transgenic crops in India and the GEAC route

A transgenic crop carries a gene from another species. India's story here is short and worth learning exactly, because the exam asks the status, not the science. Bt cotton carries cry genes from the soil bacterium *Bacillus thuringiensis*. The bacterium makes a crystal protein that is an inactive protoxin - harmless as it sits. When a bollworm eats it, the alkaline gut of the insect activates the toxin, which punctures the gut lining and kills the larva. The human stomach is acidic, so the protein is not activated in us. That mechanism is the reason Bt is considered selective.

GM and gene-edited crops in India

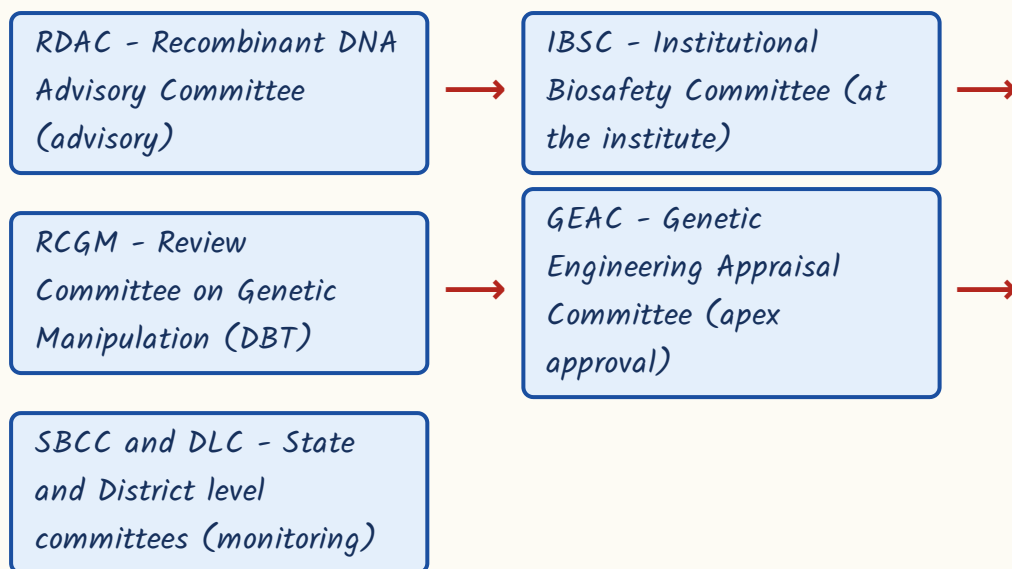


- Bt cotton carries cry genes - cryIAc and cry2Ab - against bollworms.
- The Bt protein is an inactive protoxin crystal and becomes active only in the alkaline gut of the insect.
- Bollgard-I Bt cotton was approved in India in March 2002.
- Bt cotton remains the **ONLY** GM crop approved for commercial cultivation in India.
- Golden Rice is engineered to make beta-carotene (provitamin A) in the endosperm; Flavr Savr tomato in 1994 was the first genetically modified food crop to be commercialised anywhere.

The three Indian GM crop cases - learn the status, not just the science

Crop	Gene / technology	Indian status
Bt cotton (Bollgard-I)	cryIAc from <i>Bacillus thuringiensis</i>	Approved March 2002; the only GM crop grown commercially in India
Bt brinjal	cryIAc, developed by Mahyco	GEAC cleared it in 2009; moratorium imposed February 2010 by Environment Minister Jairam Ramesh. Bangladesh grows it commercially
GM mustard DMH-II	Barnase-barstar-bar system, Deepak Pental's team, University of Delhi	GEAC recommended environmental release October 2022; Supreme Court split verdict July 2024, referred to a larger bench

India's GMO regulatory chain under the 1989 Rules



- *GMOs in India are regulated under the 1989 Rules framed under the Environment (Protection) Act, 1986.*
- *GEAC - the Genetic Engineering Appraisal Committee - sits under the Ministry of Environment, Forest and Climate Change and is the apex body for approving environmental release of GMOs.*
- *The Biological Diversity Act, 2002 created the National Biodiversity Authority at Chennai and the State Biodiversity Boards.*
- *At the local level it created Biodiversity Management Committees, which prepare People's Biodiversity Registers.*
- *The Biological Diversity (Amendment) Act, 2023 decriminalised offences and exempted registered AYUSH practitioners from benefit sharing.*
- *India is a party to the Cartagena Protocol on biosafety and the Nagoya Protocol on access and benefit sharing.*

TRAP

Bt cotton is the only GM crop approved for COMMERCIAL CULTIVATION in India - that is the phrase that matters. Bt brinjal was cleared by GEAC and then stopped by a moratorium; GM mustard was recommended by GEAC but is stuck in court. Cleared, recommended and approved for cultivation are three different things, and the paper uses that difference.

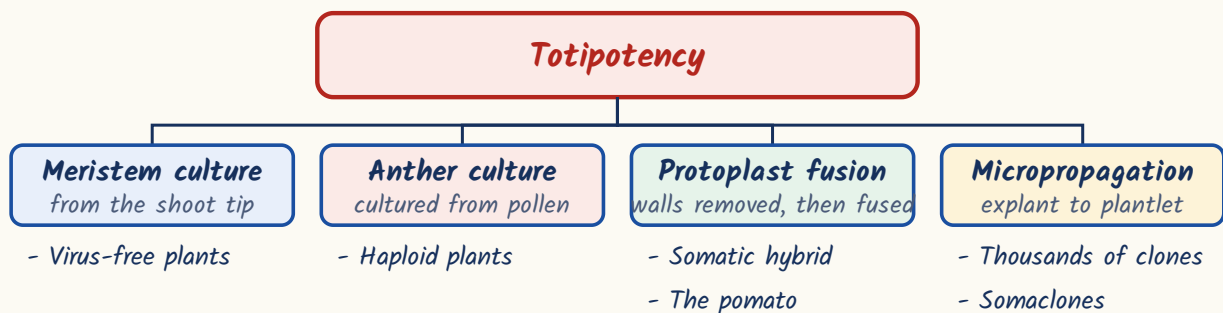
CURRENT LINK

Current link (as of September 2026): the GM mustard DMH-11 case remains unresolved. The Supreme Court's two judges split in July 2024 and the matter went to a larger bench, so no new GM food crop has been released for cultivation. If any movement happens before the exam, the fact to update is only this one line - GEAC recommendation October 2022, split verdict July 2024, status thereafter.

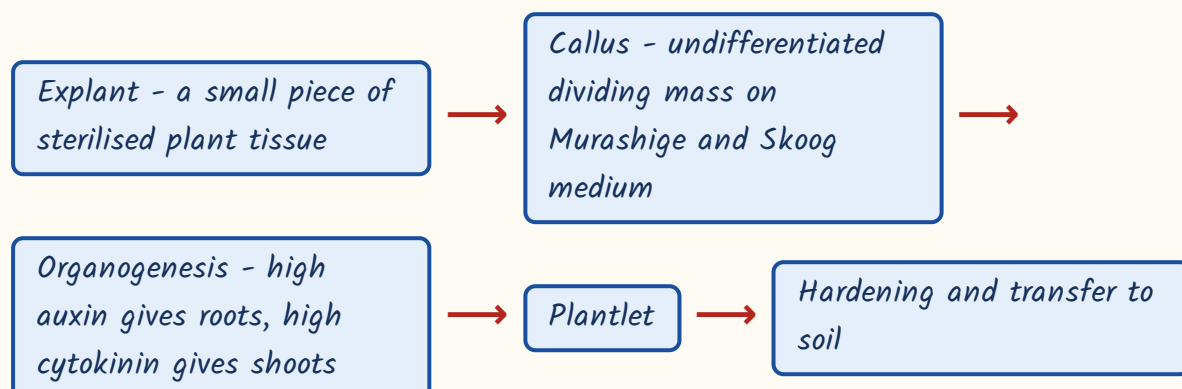
Tissue culture, biofertilisers and industrial biotechnology

Plant tissue culture rests on one property - totipotency. Any living plant cell carries the full genetic instruction set, so a single cell can be coaxed into rebuilding the whole plant. You take a small piece of tissue, the explant, put it on a nutrient medium under sterile conditions, and it first becomes an unorganised mass of dividing cells, the callus. From there, the balance of two hormones decides what grows: more auxin gives roots, more cytokinin gives shoots. That single ratio is the fact the exam asks.

What each tissue culture route gives you



Micropropagation - from a piece of tissue to a whole plant



- G. Haberlandt is the Father of plant tissue culture; the underlying principle is totipotency.
- Murashige and Skoog (MS) medium is the standard culture medium.
- Meristem or shoot-tip culture gives virus-free plants, because the meristem is usually free of virus.
- Anther culture gives haploid plants; protoplast fusion (somatic hybridisation) produced the 'pomato' from potato and tomato.
- Micropropagation produces thousands of genetically identical plants - somaclones - from one parent.
- In nitrogen fixation the enzyme nitrogenase does the work, and leghaemoglobin in the root nodule protects it from oxygen, which would destroy it.
- Bioremediation - *Pseudomonas putida*, the oil-eating 'superbug' engineered by Ananda Mohan Chakrabarty, became the first living organism to be patented, in *Diamond v. Chakrabarty*, 1980.

Biofertilisers and biopesticides - match the organism to the job

Organism	Type	What it does
<i>Rhizobium</i>	Biofertiliser	Symbiotic nitrogen fixation in legume root nodules
<i>Azotobacter</i>	Biofertiliser	Free-living aerobic nitrogen fixer
<i>Azospirillum</i>	Biofertiliser	Associative nitrogen fixer
<i>Anabaena</i> , <i>Nostoc</i> ; <i>Azolla</i> with <i>Anabaena azollae</i>	Biofertiliser	Cyanobacterial nitrogen fixation, used in paddy
<i>Frankia</i>	Biofertiliser	Nitrogen fixation with <i>Casuarina</i>
<i>Mycorrhiza</i> / VAM	Biofertiliser	Improves phosphorus uptake
<i>Bacillus thuringiensis</i> , <i>Trichoderma</i> , <i>Beauveria bassiana</i>	Biopesticide	Bacterial and fungal control of pests and plant disease
NPV (Nucleopolyhedrovirus), <i>Trichogramma</i> , neem azadirachtin	Biopesticide	NPV is highly species specific; <i>Trichogramma</i> is a parasitic wasp

Fermentation products and the microbe behind each

Product	Microorganism
Ethanol	<i>Saccharomyces cerevisiae</i>
Citric acid	<i>Aspergillus niger</i>
Acetic acid	<i>Acetobacter aceti</i>
Penicillin	<i>Penicillium notatum</i> / <i>chrysogenum</i> - Fleming, Chain and Florey, Nobel 1945
Cyclosporin A (immunosuppressant)	<i>Trichoderma polysporum</i>
Statins (cholesterol lowering)	<i>Monascus purpureus</i>
Streptokinase (clot buster)	<i>Streptococcus</i>
Holes in Swiss cheese	<i>Propionibacterium shermanii</i>

Biofuels - the 1G to 4G ladder RPSC keeps coming back to

The generations of biofuel are not about the fuel produced - they are about what the fuel was made FROM. First generation uses food crops, which is why it competes with the food plate. Second generation moves to the parts we do not eat - crop residue, straw, woody biomass, all lignocellulosic. Third generation moves off land entirely and uses algae, which need no farmland. Fourth generation goes one step further and engineers the algae or microbe itself so that the process can pull in more carbon dioxide than the fuel releases - the aim is a carbon-negative fuel. Food, waste, algae, engineered algae. That is the whole ladder.

Biofuel generations - the feedstock is the answer

Generation	Feedstock	Key point
1G - first generation	Edible crops - sugarcane, molasses, corn, vegetable oils	Competes with food and land

Generation	Feedstock	Key point
2G - second generation	Non-food lignocellulosic biomass and crop residues	Uses waste; India's first 2G plant is at Panipat, Haryana, on paddy straw
3G - third generation	Algal biomass	No farmland needed, high yield per area
4G - fourth generation	Genetically or metabolically engineered algae and microbes	Designed for carbon capture; aims at carbon-negative fuel

- The National Policy on Biofuels, 2018, amended in 2022, advanced the 20% ethanol blending target from 2030 to 2025-26. India achieved 20% blending in 2025.
- Pradhan Mantri JI-VAN Yojana supports second-generation ethanol projects.
- SATAT - Sustainable Alternative Towards Affordable Transportation - promotes compressed biogas.
- Biodiesel comes from transesterification of vegetable oils; jatropha was the crop pushed for it in India.

ASKED IN RAS

Asked in RAS Pre 2013 and again in RAS Pre 2024 - first-generation biofuels are those produced from edible food crops such as sugarcane, corn and vegetable oils. A topic asked eleven years apart and then repeated is a topic that is part of the RPSC template.

ASKED IN RAS

Asked in RAS Pre 2013 and 2024 - biofuels from algal biomass are third generation, and fourth-generation biofuels are distinguished by the use of genetically or metabolically engineered algae and microbes aimed at capturing carbon dioxide. If the question names the feedstock, count up the ladder; if it names 'engineered', the answer is 4G.

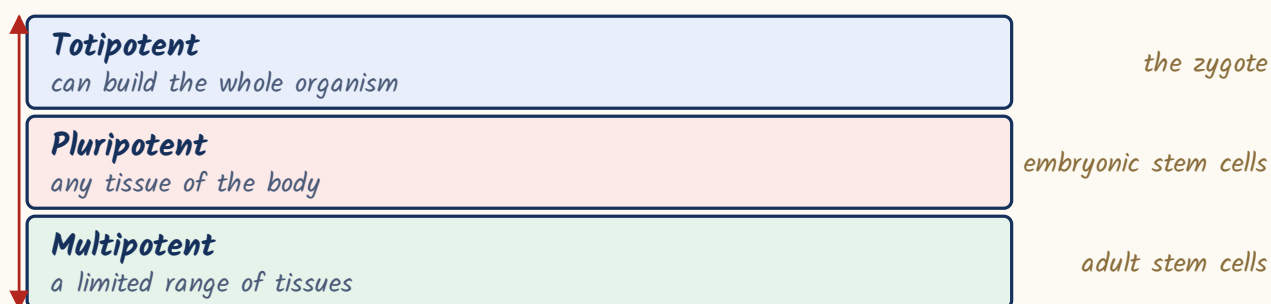
ASKED IN RAS

Asked in RAS Pre 2024 as a statement set - India's first second-generation ethanol bio-refinery is at Panipat in Haryana and runs on paddy straw, and the 20% blending target was advanced to 2025-26 by the amended National Policy on Biofuels. Learn the place name; that is the part candidates miss.

Cloning, stem cells, gene therapy, vaccines and the Indian set-up

The medical end of biotechnology is a name-and-date section, so treat it as one. Cloning made its point with Dolly - the first mammal produced from an adult body cell, which proved that a fully specialised cell still holds the complete instruction set. Stem cells work from the other direction: cells that have not yet specialised, graded by how many things they can still become. Gene therapy tries to fix the fault at source by supplying a working copy of the gene. And the COVID years handed the exam a ready-made matching set of vaccine platforms, which is now standard material.

Stem cells - potency falls down the tiers



- Dolly the sheep, born 5 July 1996 at the Roslin Institute to Ian Wilmut and Keith Campbell, was the first mammal cloned from an adult somatic cell.
 - The technique was somatic cell nuclear transfer (SCNT). India's cloned buffalo work was done at NDRI, Karnal.
 - Stem cell grades - totipotent is the zygote, which can make the whole organism plus the placenta; pluripotent are embryonic stem cells, which can make any body tissue.
 - Multipotent are adult stem cells, with a limited range of possible tissues.
 - Shinya Yamanaka shared the 2012 Nobel Prize with John Gurdon for induced pluripotent stem cells - adult cells reprogrammed back to a pluripotent state.
 - Umbilical cord blood is the common practical source of stem cells for banking.
-
- Gene therapy was first attempted in 1990 on a child with adenosine deaminase (ADA) deficiency.
 - Somatic gene therapy changes body cells only; germline changes would pass to the next generation and are not permitted.
 - RNA interference (RNAi), used for pest resistance in plants, earned Andrew Fire and Craig Mello the 2006 Nobel Prize.
 - Monoclonal antibodies are produced by hybridoma technology - Kohler and Milstein, Nobel 1984 - and their names carry the suffix '-mab'.

COVID vaccine platforms - a ready-made matching set

Vaccine	Platform	Maker
Covaxin (BBV152)	Whole-virion inactivated	Bharat Biotech with ICMR-NIV Pune
Covishield	Chimpanzee adenovirus viral vector (Oxford-AstraZeneca)	Serum Institute of India
ZyCoV-D	DNA plasmid - world's first DNA vaccine approved for human use, three needle-free doses	Zydus
GEMCOVAC-19	mRNA - India's first indigenous mRNA vaccine	Gennova

Vaccine	Platform	Maker
Corbevax	Recombinant protein subunit	Biological E

- The Department of Biotechnology (DBT) was set up in 1986 under the Ministry of Science and Technology.
- BIRAC - Biotechnology Industry Research Assistance Council - was set up in 2012 to support biotech start-ups and small firms.
- The BioE3 Policy - Biotechnology for Economy, Environment and Employment - was approved by the Union Cabinet in August 2024 to promote high-performance biomanufacturing and biofoundries.
- Remember the division of labour: DBT funds and promotes, BIRAC backs start-ups, GEAC approves environmental release, and the National Biodiversity Authority handles access and benefit sharing.

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Fix the four Indian COVID vaccines by platform, because that is how the matching question is set - Covaxin is inactivated, Covishield is a viral vector, ZyCoV-D is DNA and a world first, GEMCOVAC is mRNA and an Indian first. Two 'firsts' in one row is exactly what the paper likes.

REVISION RECAP

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5. Ti plasmid of *Agrobacterium tumefaciens*, the crown gall pathogen, is the commonest vector for dicot plants.

6. Gene transfer - microinjection for animal cells, gene gun for plant cells, electroporation, calcium treatment for bacteria.
7. Human insulin (Humulin, 1982-83) was the first commercial rDNA product - A chain 21 and B chain 30 amino acids joined by disulphide bonds.
8. PCR - Kary Mullis, Nobel 1993; denature 94-95 C, anneal 50-55 C, extend 72 C with Taq polymerase from *Thermus aquaticus*.
9. In gel electrophoresis DNA moves to the anode, is stained with ethidium bromide and seen under UV. Southern DNA, Northern RNA, Western protein.
10. CRISPR-Cas9 comes from a bacterial antiviral defence, is guided by a guide RNA; Charpentier and Doudna, Nobel 2020; FELUDA by CSIR-IGIB; Casgevy the first approved CRISPR therapy.
11. Bt cotton carries cry1Ac and cry2Ab; the protoxin activates only in the insect's alkaline gut; approved March 2002 and still India's only commercially cultivated GM crop.
12. Bt brinjal - GEAC cleared 2009, moratorium February 2010 by Jairam Ramesh; Bangladesh grows it. GM mustard DMH-11 - barnase-barstar-bar, GEAC October 2022, Supreme Court split verdict July 2024.
13. GMO chain in India - RDAC, IBSC, RCGM, GEAC, SBCC, DLC, under the 1989 Rules of the Environment (Protection) Act 1986; GEAC sits under MoEFCC.
14. Tissue culture rests on totipotency, Haberlandt is its father; explant to callus to plantlet on MS medium; high auxin gives roots, high cytokinin gives shoots.
15. Meristem culture gives virus-free plants, anther culture gives haploids, protoplast fusion gave the pomato.
16. Biofertilisers - Rhizobium symbiotic, Azotobacter free-living, Azospirillum associative, Azolla-Anabaena in paddy, mycorrhiza for phosphorus.
17. Biofuels - 1G edible crops, 2G lignocellulosic residue, 3G algae, 4G engineered algae for carbon capture; India's first 2G plant at Panipat on paddy straw; 20% blending achieved 2025.
18. Dolly 1996 by SCNT at the Roslin Institute; stem cells totipotent, pluripotent, multipotent; iPSC by Yamanaka with Gurdon, Nobel 2012.
19. Gene therapy first attempted 1990 for ADA deficiency; RNAi Nobel 2006 to Fire and Mello; monoclonal antibodies by hybridoma, suffix '-mab'.
20. Covaxin inactivated, Covishield viral vector, ZyCoV-D the world's first human DNA vaccine, GEMCOVAC India's first mRNA vaccine; DBT 1986, BIRAC 2012, BioE3 Policy August 2024.